

Wallet-Level Markout on Hyperliquid: Persistence and Copyability

1. Research question and summary

Can wallet-level markout identify Hyperliquid traders whose entries contain repeatable and copyable information?

We measure post-fill markout for taker entries on BTC, ETH, SOL and HYPE over August 2025 – June 2026, rank wallets by markout, test whether high-ranked wallets persist out-of-sample, and characterise the wallets that recur.

Three findings:

- Static full-history rankings did not identify a durable top-wallet cohort: wallets ranked on the training period showed no reliable relationship to validation-period markout.
- Rolling monthly ranks showed modest one-month-ahead persistence: the prior-month top decile outperformed the field by approximately 4.4 bp at the 4-hour horizon and 11.4 bp at the 8-hour horizon (neutralized, wallet-day-weighted).
- Recurring high-ranked wallets were predominantly contrarian, entering after larger, decelerating price moves. Their returns were close to those of a mechanical reversal benchmark evaluated at the same timestamps; we found no statistically resolved incremental return attributable to wallet identity or direction, though the sample cannot exclude a smaller day-capped timing effect.

2. Data and methodology

Universe and dates. 7,332,013 taker entries from 32,949 wallets across 273 instruments, 2025-08-01 to 2026-06-30 (UTC). The analysis is restricted to BTC, ETH, SOL and HYPE.

Cohort. To study copyable behaviour we freeze a mid-frequency taker cohort, selected on identity rather than performance: ≥ 200 majors fills, taker share ≥ 0.70 , median round-trip hold 1–24 h, and ≥ 200 training entries. This yields **1,694 wallets and 812,616 priced entries** (funnel: 29,334 wallets with ≥ 200 fills $\rightarrow$ 12,544 copyable takers $\rightarrow$ 1,694).

Splits. Training < 2026-02-01 (583,032 entries); embargo February 2026 (61,116, discarded); validation 2026-03-01 to 06-30 (168,468). The validation window is a -10.85% BTC period.

Observation unit. One observation is a position-increasing taker decision: a (wallet, coin, bar, direction) cell. A fill is scored only if it is taker-crossed, non-wash, and increases absolute position; maker, wash and position-reducing fills are tracked for position accounting but not scored. Same-bar, same-direction opening fills are aggregated into one entry at the shared next-bar price. Scaling into a position across several bars therefore produces several entries, each weighted once.

Markout. For horizon h , $\text{markout} = \text{dir} \cdot (\text{close}(\text{first bar after entry} + h) / \text{close}(\text{first bar after entry}) - 1)$, in basis points. Both endpoints use the first bar strictly after the timestamp, so pricing is post-fill. This is a hypothetical gross signed price return; it excludes spread, fees, funding and any modelled exit, and is not realized PnL. Horizons: 1, 2, 4, 8, 24 h.

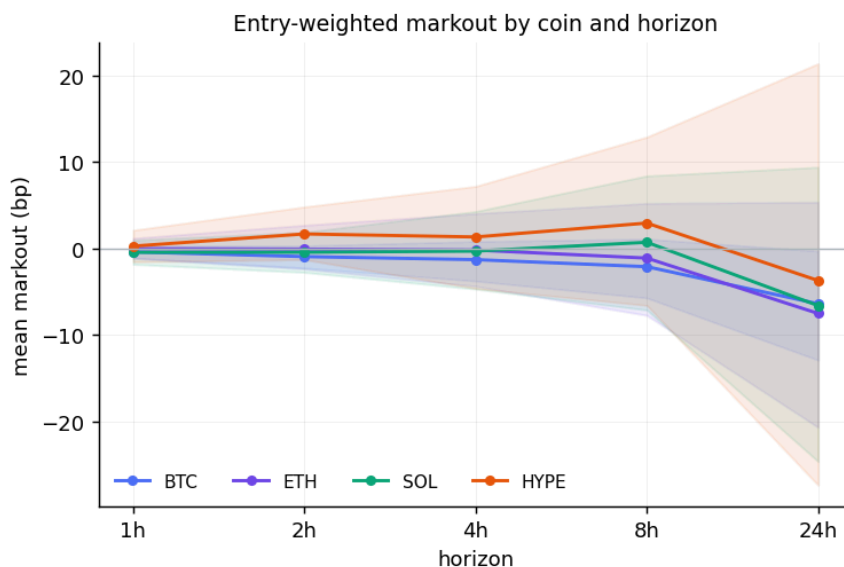
Weighting. Results are labelled by weighting: term-structure and top-wallet tables are entry-weighted; the wallet-skill comparison is wallet-equal-weighted; persistence and decile tests are wallet-day-weighted; deployability estimates are wallet-day-capped (equal capital per wallet-day).

Inference. Confidence intervals use a calendar-day block bootstrap, which preserves the dependence induced by overlapping forward windows and same-day cross-wallet correlation. Selection tests across many wallets/metrics are controlled with Benjamini–Hochberg FDR and, where a single best cell is reported, a Romano–Wolf maximum-statistic null. Cost assumptions use a placeholder 6–9 bp per-coin round-trip, treated as an upper bound pending order-book calibration.

Limitation. The March–June validation window was examined repeatedly during the study and informed later specification choices (cohort horizon, cost model, control design). Validation results are therefore historical / exploratory out-of-sample; only prospective data can provide clean confirmation.

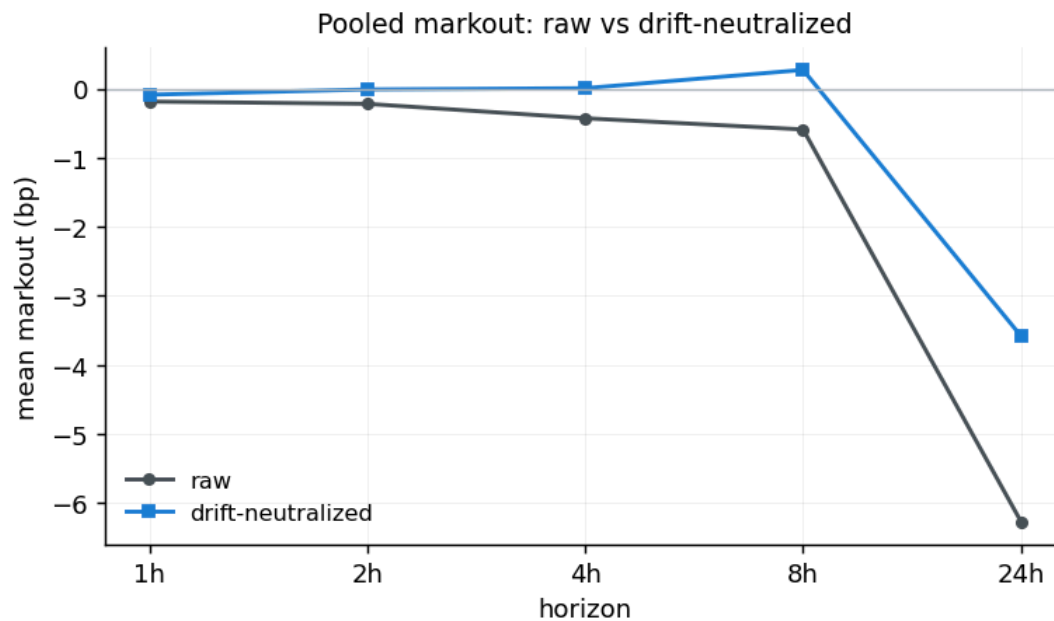
3. Baseline markout results

Entry-weighted markout by coin and horizon is small relative to its dispersion. Pooled across coins, the mean is -0.19 bp at 1 h and -0.60 bp at 8 h, with day-block intervals spanning zero at every horizon through 8 h; only the 24-h horizon is materially negative (-6.3 bp pooled; BTC -6.4 bp, interval excluding zero). HYPE is the only coin with a positive short-horizon mean (about $+3$ bp at 8 h), but its interval spans zero and its 24-h raw mean ($+2.8$ bp) reverses to -18.6 bp once the coin’s own monthly drift is removed. Most of the visible coin-level structure is within sampling noise at the day-block level.



Full sample, entry-weighted. Bands: 95% calendar-day block bootstrap. Markout = post-fill signed price return to horizon (gross, no costs).

Neutralising each coin’s monthly mean forward return removes most of the long-horizon movement, indicating that the raw 24-h figures are largely coin drift rather than entry-timing information.



Pooled over four coins, entry-weighted. Neutralized = raw minus each coin's own monthly mean forward return.

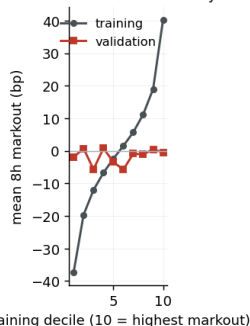
Dispersion grows with horizon (pooled standard deviation ≈ 93 bp at 1 h to ≈ 363 bp at 24 h); per-coin quantiles are tabulated in the appendix.

4. Wallet-rank persistence

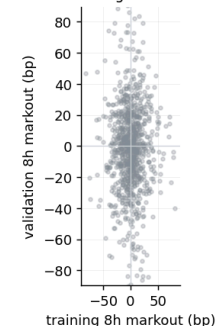
We separate two analyses that answer different questions.

Static ranking. Wallets are ranked once on the training period and measured in validation. The relationship is weak: the train-to-validation rank correlation of mean 8-hour markout is approximately +0.02 (not significant, $n \approx 990$), and the highest training deciles do not retain an advantage in validation. Selecting wallets on best in-sample markout produces -28.7 bp net in validation. A formal walk-forward selection test across nine ranking metrics and thousands of cells returned a best coin p-value of 0.09 with no survivors after FDR control.

Training and validation 8-hour markout by training decile

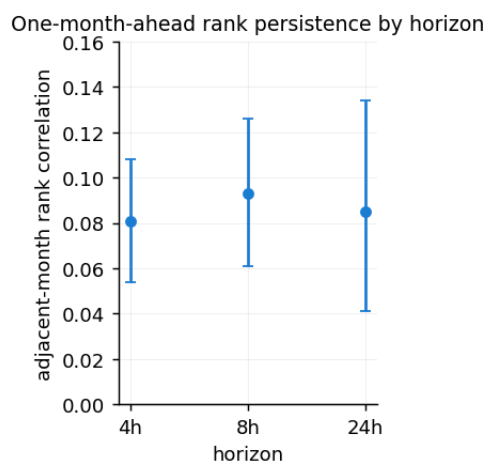


Validation vs training wallet 8-hour markout



Static ranking: wallets ranked once on training, measured in validation. Entry-weighted per wallet. Train-validation rank correlation $\rho = +0.02$ ($p = 0.59$), $n = 992$.

Rolling ranking. Wallets are re-ranked each month and evaluated one month ahead. Here a modest signal is present: the adjacent-month rank correlation of neutralized, wallet-day-weighted markout is +0.093 at 8 h (95% interval [+0.061, +0.126]; positive in 10 of 10 folds), with similar values at 4 h (+0.081) and 24 h (+0.085). The prior-month top decile outperformed the field by +4.4 bp at 4 h ([+1.8, +6.8]) and +11.4 bp at 8 h ([+5.3, +17.0]).



Rolling monthly re-ranking (wallet-day-weighted, drift-neutralized). Spearman rank IC of month t rank vs month $t+1$ markout; 95% fold bootstrap.

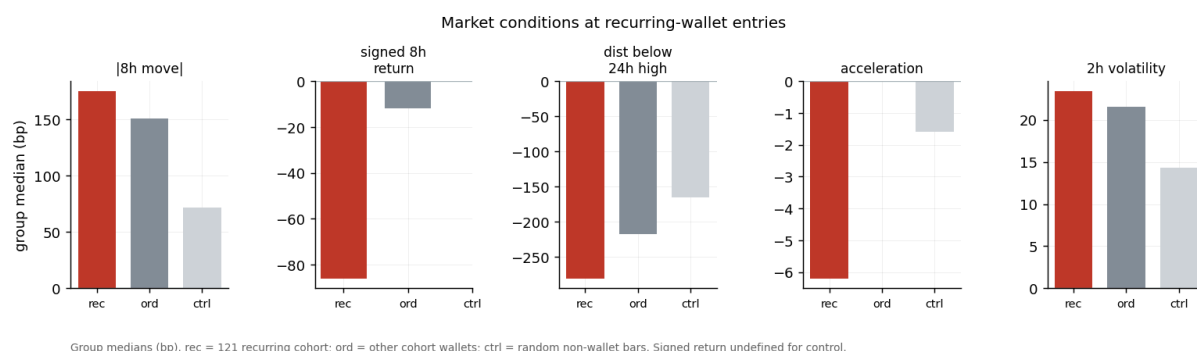
The two analyses lead to different conclusions and should not be combined: static full-history selection did not generalise, whereas the rolling cross-sectional ranking exhibited modest one-month-ahead predictive value.

Requiring recurrence sharpens this. Counting how often a wallet appears in the monthly top decile against a binomial chance benchmark: appearance in ≥ 3 months (48 wallets observed vs 35 expected, $p = 0.017$) exceeds chance; the ≥ 2 -month (204 vs 184, $p = 0.059$) and ≥ 4 -month (9 vs 5, $p = 0.066$) counts do not clear the 0.05 threshold.

5. Behavioral and benchmark decomposition

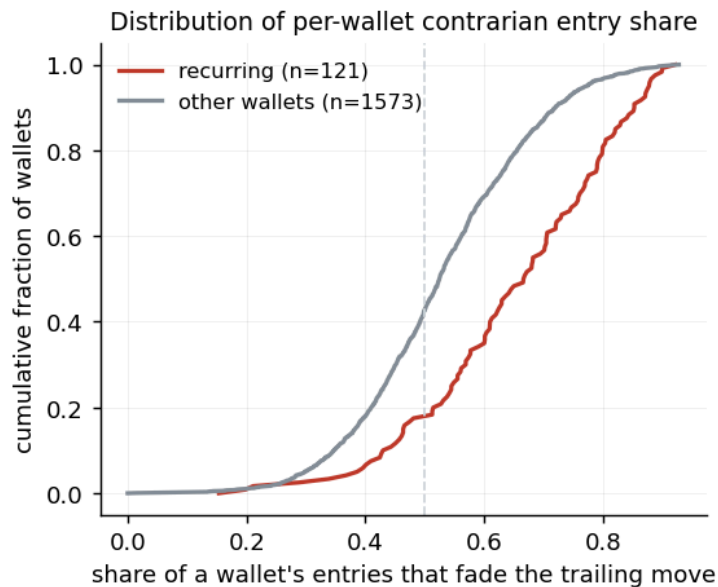
Recurring cohort. We freeze a recurring cohort as wallets in the top decile in at least two of the six training months (121 wallets), using drift-neutralized 8-hour markout and wallet-day weighting. The two-month threshold was chosen for breadth and statistical power (121 wallets versus 28 at a three-month threshold); only the three-month subset independently cleared the chance benchmark above, so the 121 should be read as a broad recurring set rather than 121 individually established repeat performers. What distinguishes the cohort is conviction and cross-coin breadth rather than size or activity: mean training rank percentile 0.72 versus 0.48 for the field, activity in all four coins versus three, near-identical median ticket (about \$4.1k), and a modest long bias.

Entry conditions. Recurring-cohort entries occur after larger trailing moves and deeper pullbacks than other wallets or random bars, and against the direction of the preceding move.



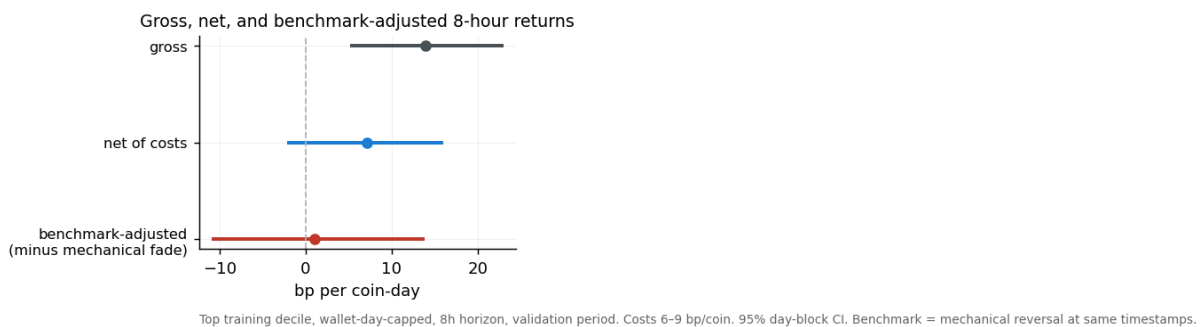
Group medians (bp). rec = 121 recurring cohort; ord = other cohort wallets; ctrl = random non-wallet bars. Signed return undefined for control.

At the wallet level this contrarian tendency is common but not universal: 83% of the 121 have a net-contrarian entry record and 65% fade the trailing move on at least 60% of entries, versus a median contrarian share of 0.67 for the cohort against 0.52 for other wallets.



Per wallet, ≥ 20 training entries. Contrarian = entry direction opposite the trailing 8h move. Dashed line = 50% (coin-flip).

Benchmark comparison. We compare the high-markout wallets' returns to a mechanical reversal benchmark (`fade_dir = -sign(trailing 8-hour move)`) evaluated at the same timestamps, so execution effects cancel in the difference. For the training-ranked top decile at 8 h, gross return is +14.0 bp ([+5.4, +22.8]) and net of costs +7.1 bp; the benchmark-adjusted return is +1.0 bp ([−10.75, +13.66], $p = 0.44$). Most of the estimated advantage is captured by the mechanical benchmark.

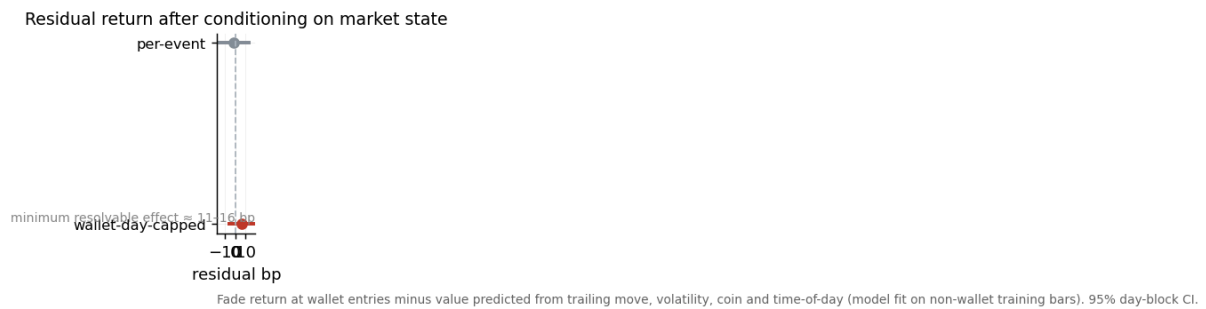


Decomposing the timing question: trading in the wallet's direction did not improve on the fade at the same times (A versus B, difference not resolved), the unconditional fade is negative net of costs (C = −7.0 bp), and the fade at activity-matched non-wallet times is also negative (D = −6.4 bp). Wallet direction did not improve performance in this specification.



Conditioning the fade return on observable market state (trailing move, volatility, coin, time of day), fit on non-wallet training bars and applied to wallet entries, leaves a per-event residual near zero (−1.3 bp) and a wallet-day-capped residual of +5.9 bp ([−5.5, +17.7]). The minimum effect this validation sample can resolve is approximately 11–16 bp, so a smaller incremental timing effect cannot be excluded, but none is

established. A nested return model confirms the direction of this result: adding wallet identity, rank, direction and size to a market-state model changed cross-fit predictive correlation by -0.0003 .



In summary: the recurring cohort’s entry returns are consistent with a short-horizon reversal pattern; observable market-state variables explained most of the estimated advantage; no incremental return from wallet identity or direction was statistically resolved; a smaller day-capped timing effect remains possible within the confidence interval and the sample’s resolution.

6. Conclusion and forward test

Static rankings of wallet-level markout did not generalise reliably from the training period to the historical validation period. Rolling monthly ranks exhibited modest one-month-ahead persistence, with the prior-month top decile outperforming the field by approximately 4.4 bp at 4 h and 11.4 bp at 8 h (neutralized, wallet-day-weighted).

Recurring high-ranked wallets were predominantly contrarian and tended to enter after larger, decelerating price moves. Their estimated returns were similar to those of a mechanical reversal benchmark evaluated at the same timestamps. We therefore find no statistically resolved incremental return attributable to wallet identity or direction, although the available sample cannot exclude a smaller day-capped timing effect.

These findings motivate a prospective comparison of a market-state reversal model against the same model augmented with wallet information. The prospective test will compare, on new data, a rule driven by the market-state model alone, the same rule augmented with recurring-wallet activity and direction, and direct copying of wallet direction, under identical capital constraints and a fixed 8-hour horizon, with execution, spread, fees and funding logged. Because the historical validation window informed subsequent specifications, its results are treated as exploratory; forward data is required for confirmation.

Appendix

A. Methodology controls

Point estimates are entry-weighted unless labelled; inference uses a calendar-day block bootstrap to handle overlapping forward windows and same-day cross-wallet correlation. Selection across metrics is FDR-controlled. Pricing is strictly post-fill (next bar after entry) on fine candles; an earlier hourly-candle pricing scheme was found to credit entries with pre-fill price movement and was replaced. During the study we identified and corrected five pipeline issues, listed in Section C.

B. Trade size and dispersion

Per-coin trade-size quantiles and 8-hour markout dispersion are in `out/appendix_tables.md` (trade size median $\approx$ \$4k with a heavy right tail past \$1M; 8-hour markout standard deviation 130–330 bp by coin). Regime composition, hold-time distribution and monthly volume are in `figs/03_regime.png`, `figs/03_holdtime.png`, `figs/03_temporal.png`.

C. Corrections made during the study

Item	Before → after
Test-period activity used in wallet selection	removed; train-only selection with eval-time attrition (99 → 170 wallets)
Test-fitted matched-control bins	refit on training data only
Shrinkage estimator degenerate in 4 of 11 months	flagged 100% of wallets as top-decile; corrected to a raw-rank fallback; recurrence at ≥ 3 months 686 → 48, transition $3.2\times \rightarrow 1.09\times$ (not significant)
No positive-control power check	added; validation window resolvable effect $\approx 11\text{--}16$ bp
Three inconsistent top-decile definitions	consolidated to one train-only recurring set, frozen to disk

D. Top training-decile wallets (in-sample reference)

Top wallets by training 8-hour markout, with per-trade Sharpe (mean/standard-deviation of markout, unannualized), hit rate, and maximum drawdown computed on a non-overlapping (≥ 8 -hour-spaced) entry subset. These are in-sample descriptive values; Section 4 shows they do not persist.

#	wallet	N	mean 8h (bp)	Sharpe	hit%	MDD (bp)	med \$
1	0x5aadb434...	214	+132.8	0.36	69	1338	12,061
2	0xfd490cf8...	236	+124.2	0.65	76	1676	131,053
3	0xdfb34089...	248	+83.3	0.33	64	1400	5,535
4	0x172f5a11...	202	+80.2	0.16	51	2917	3,225
5	0x1ecc7c22...	200	+78.9	0.29	61	802	25,028

Drawdown on the raw overlapping-horizon markout series is not reported because it scales with trade cadence rather than risk.

E. Number-to-source map

See `docs/NUMBER_SOURCE_MAP.md` for each headline estimate mapped to its generating script and output file.